Sunil Parameswaran

AI Engineer | Machine Learning Engineer | Data Scientist
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PROFESSIONAL SUMMARY

AI/ML engineer and MSc Artificial Intelligence student (Leiden University) who has taken multilingual, multimodal LLM systems to production for **1,000+ concurrent users with zero downtime**. Strengths: retrieval-augmented generation (RAG), LLM fine-tuning, behavior modeling and evaluation/observability. Author of 3 peer-reviewed publications.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript/Node.js, C, Bash
AI/ML: LLMs, RAG, fine-tuning, NLP, deep learning, Bayesian modeling, behavioral classification, recommendation systems, AutoML, agent evaluation
Frameworks & Platforms: PyTorch, TensorFlow, Keras, scikit-learn, OpenAI API, Langfuse, Flask, React, PostgreSQL, vector search, Docker, Kubernetes, AWS EC2, NGINX

PROFESSIONAL EXPERIENCE

Data Scientist | Quest Alliance *Aug 2025 - Jul 2026 | Bangalore, India*

- Architected and productionized **AskAbhi**, a multilingual, multimodal AI learning companion delivering employability and career guidance across **8 Indian languages**.
- Designed an end-to-end pipeline: responsible-AI pre-checks, semantic retrieval, behavior classification, Bayesian learner profiling, adaptive nudging, grounded RAG generation and post-generation safety checks; reported **92% context relevance** and **71% classification accuracy**.
- Fine-tuned **GPT-4.1 nano** on domain- and behavior-specific data; built a Dirichlet learner model over **33 micro-behaviors** with temporal evidence updates and effective sample size.
- Deployed with Node.js, React, PostgreSQL, vector retrieval, Langfuse, AWS EC2, NGINX and async processing; built evaluation and observability pipelines.

Data Science Intern | Quest Alliance *Jun 2024 - Aug 2024; Jan 2025 - Jul 2025 | Bangalore, India*

- Built the original RAG proof of concept and early multimodal (text, speech, image) architecture for AskAbhi; improved grounding via annotation, multilingual translation, retrieval tuning, evaluation and field testing.
- Prototyped an AI-powered issue-management assistant for support triage and ticket routing.

University Teaching Assistant, Generative AI | PES University *Jan 2025 - May 2025 | Bangalore, India*

- Mentored students building generative AI applications, developed course material and evaluated LLM-based projects.

Research Intern, Deep Learning | IEEE Computer Society *Apr 2023 - Sep 2023 | Bangalore, India*

- Led a four-member team building a facial emotion recognition CNN in TensorFlow/Keras; achieved **80% test accuracy on FER2013+**; work published in two peer-reviewed papers.

SELECTED PROJECTS

EmoHealth - Multimodal Deep Learning Capstone *2024 - 2025 | Best Capstone Project among 210 teams*

- Built a real-time emotion and health analytics system fusing speech, physiological signals and NLP using TensorFlow, VGGish and CoreML (on-device inference).

Agent Reasoning with BALROG *Leiden University | Group project*

- Evaluated baseline, persistent-planning and bounded actor-critic reflection LLM agents on BabyAI using Qwen and Gemini; analyzed **150 episodes**, invalid actions and search trajectories.

Second Opinion - Wastewater Measurement Review *Group project*

- Contributed to an explainable anomaly-detection prototype using rolling historical baselines, an experimental Isolation Forest and an interactive dashboard to prioritize unusual wastewater measurements.

LLM From Scratch *Independent project*

- Trained BPE tokenizers and explored pretraining, reasoning adaptation and instruction fine-tuning of a **12.98M-parameter transformer** built on nanochat.

EDUCATION

M.Sc. Computer Science - Artificial Intelligence | Leiden University *Sep 2026 - Present | Leiden, Netherlands*

B.Tech CSE (Machine Intelligence & Data Science) | PES University *2021 - 2025 | CGPA 8.03/10*

PUBLICATIONS

- EmoHealth: A Deep Learning Fusion of Speech and Body Parameters for Real-Time Emotion Detection and Health Analytics - Springer, ICTIS 2025.
- Deep Convolutional Neural Network Architecture for Facial Emotion Recognition - PeerJ Computer Science, 2024.
- Comparative Analysis of Facial Expression Recognition Algorithms - Springer LNNS, 2024.

HONORS & LEADERSHIP

Best Capstone Project: 1st among 210 teams, PES University. **Hackathon technical mentor:** mentored teams placing 1st and 3rd. Selected as 1 of 12 classical guitarists globally for the Guitar Foundation of America International Mentorship Program.